

AWS EFS Lab: Multi-AZ Setup with EC2 and Mounting

Objective:

- Launch 2 EC2 instances in **different AZs**.
 - Create **EFS** and mount it on both instances.
 - Configure **security groups** to allow **NFS (port 2049)**.
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Prerequisites:

- AWS account
 - VPC with at least 2 subnets in different AZs
 - IAM role with EC2 and EFS access (optional but preferred)
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Step 1: Create EFS File System

1. Go to **EFS Console** → **Create file system**
2. Select your **VPC**
3. Enable **automatic mount targets** for **both AZs** (select 2 subnets)
4. Enable encryption (default)
5. Create the file system

 **Note the EFS File System ID:** `fs-xxxxxxx`

Step 2: Configure Security Group for EFS

Create a new EFS-SG:

- Inbound Rule:
 - **Type:** NFS
 - **Protocol:** TCP
 - **Port:** 2049
 - **Source:** Security group of your EC2 instances (e.g., `EC2-SG`)

Example:

Type	Protocol	Port	Source
NFS	TCP	2049	<code>EC2-SG</code>

Apply **this SG** to all EFS mount targets.

Step 3: Launch EC2 Instances (2x)

- Launch two EC2 instances (Amazon Linux 2 preferred)
- Place them in **different subnets/AZs** (e.g., `us-east-1a` and `us-east-1b`)
- Assign them the **same EC2-SG** security group
- Ensure they can connect to each other and to EFS (use same VPC)

#Once the instances are launched

#Connect the Elastic File System with the Servers

#Before we try to connect

#We need to configure security-groups for the EFS

- Go to VPC
- Go to SG
- Create SG
- Inbound rule - 2049

#Once the SG is created

- Go to EFS
- Select file-sys
- Go to network
- Manage network
- Delete default SG
- Add efs-nfs SG

#Once the SG is configured, Now efs can be connected to the Vm's

#To connect efs

- Go to EFS
- Go to attach
- Copy EFS address

#Go to vm

- Create a folder - efs
- Mount the EFS with folder

#Once the EFS is mounted, the data stored on efs folder will be replicated across all ec2 instances where EFS is connected.